Probability and Statistics 1

Tutorial Sheet — Session 3

Position parameters: mode, arithmetic, weighted, geometric, harmonic and quadratic means, median, quartiles and interquartile range

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Exercise 1

The number of commits pushed per day was recorded on a sample of $n = 50$ repositories of an organisation. The results are:

Commits x_i	0	1	2	3	4	5
n_i	8	14	12	9	5	2

(a) What is the mode of the distribution? Justify.

(b) Compute the increasing cumulative absolute frequencies $n_i^{\nearrow}$.

(c) Compute the arithmetic mean $\bar{X}$.

(d) Compute the median Me, the first quartile Q_1 and the third quartile Q_3, then the interquartile range $IQR = Q_3 - Q_1$.

(e) Compare the mean and the median. What does their difference tell you about the shape of the distribution?

Exercise 2

(a) A three-class classifier is evaluated on a test set of 2000 examples:

Class	Number of examples	Accuracy on the class
Class 1	1200	0.98
Class 2	300	0.85
Class 3	500	0.70

Compute the unweighted mean of the three class accuracies, then the true overall accuracy of the model on the test set. Which of the two numbers should be reported? Justify.

(b) A service exposes three endpoints, monitored over one minute:

Endpoint	Requests per minute	Mean latency (ms)
`/health`	2000	4
`/search`	300	250
`/recommend`	100	120

Compute the unweighted mean of the three endpoint latencies, then the mean latency over all requests. Which of the two numbers describes the experience of a user? Justify.

Exercise 3

(a) *Geometric mean.* A compiler optimisation yields, on five benchmarks, the speedups 1.6, 1.9, 2.2, 1.4, 1.7 with respect to the unoptimised code. Compute the geometric mean G and the arithmetic mean $\bar{X}$ of the five speedups. Which of the two summarises the five ratios correctly, and why?

(b) *Harmonic mean.* Three servers process the same stream of jobs; their throughputs are 40, 120 and 60 requests per second. Each server handles the same number of requests. Compute the harmonic mean H and the arithmetic mean $\bar{X}$ of the three throughputs. Check that $H \leq \bar{X}$.

(c) *Quadratic mean.* A regression model is evaluated on five test points, on which the signed errors are 0.02, -0.01, 0.03, -0.02, 0.04. Compute the quadratic mean Q of the errors (the RMSE) and the mean of the absolute errors. Which of the two is the larger, and why?

(d) Verify on the three examples above that the chain of means $H \leq G \leq \bar{X} \leq Q$ holds.

Exercise 4

The inference latencies of 15 requests, in milliseconds, are:

$$12,\ 15,\ 11,\ 18,\ 14,\ 16,\ 13,\ 17,\ 15,\ 19,\ 12,\ 20,\ 16,\ 15,\ 22.$$

(a) Sort the series in increasing order.

(b) Compute the median Me.

(c) Compute the first and third quartiles Q_1 and Q_3, then the interquartile range IQR.

(d) Compute the mean $\bar{X}$. Compare it with the median and comment.

(e) How many observations lie in the interval $[Q_1, Q_3]$? Express this number as a percentage of the sample. Is this percentage always the same, whatever the series?

Exercise 5

The inference latency of a model was measured on $n = 100$ requests and grouped in five classes of common amplitude 10 ms:

Class	$[30;40[$	$[40;50[$	$[50;60[$	$[60;70[$	$[70;80[$
n_i	8	22	41	21	8

(a) Give the centre of each class and compute the increasing cumulative absolute frequencies.

(b) Compute the mode Mo by interpolation inside the modal class.

(c) Compute the median Me by interpolation inside the median class.

(d) Compute Q_1 and Q_3 by interpolation, then the interquartile range IQR.

(e) Compare Mo, Me and $\bar{X} = 54.9$ ms. What does their near-equality indicate about the distribution? Explain why this conclusion must be stated with caution for a grouped variable.

Exercise 6

For each situation below, name the position parameter that must be used (mode, median, arithmetic mean, weighted mean, geometric mean, harmonic mean or quadratic mean) and justify in one sentence.

(a) Summarising the dominant attack type among the incidents reported in a year.

(b) Summarising the latency of a service whose distribution has a long right tail.

(c) Summarising the average growth factor of a dataset whose size is multiplied by a random factor each month.

(d) Summarising the accuracy of a model on an imbalanced dataset.

(e) Summarising the average speed of a network link over several transfers of *equal size* performed at different speeds.

(f) Summarising the error of a regression model when the loss used for training is the squared error.